SUSTech_Beamer

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自动化与智能制造学院

SCHOOL OF AUTOMATION AND INTELLIGENT MANUFACTURING

1 Prerequisites

2 RWIP Lagrangian Modeling

3 Control and Observer Methods

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Properties of Cross Product

- Anti-commutativity: $\mathbf{a} \times \mathbf{b} = -\mathbf{b} \times \mathbf{a}$
- Distributivity: $\mathbf{a} \times (\mathbf{b} + \mathbf{c}) = \mathbf{a} \times \mathbf{b} + \mathbf{a} \times \mathbf{c}$
- Scalar multiplication: $(k\mathbf{a}) \times \mathbf{b} = k(\mathbf{a} \times \mathbf{b}) = \mathbf{a} \times (k\mathbf{b})$
- Orthogonality: $(\mathbf{a} \times \mathbf{b}) \perp \mathbf{a}$, $(\mathbf{a} \times \mathbf{b}) \perp \mathbf{b}$
- Magnitude (θ is the angle): $\|\mathbf{a} \times \mathbf{b}\| = \|\mathbf{a}\|\|\mathbf{b}\| \sin\theta$
- Geometric meaning: magnitude equals the area of parallelogram spanned by $\mathbf{a}$, $\mathbf{b}$
- **Vector triple product:** $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = \mathbf{b}(\mathbf{a} \cdot \mathbf{c}) - \mathbf{c}(\mathbf{a} \cdot \mathbf{b})$
- **Scalar triple product:** $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = \mathbf{b} \cdot (\mathbf{c} \times \mathbf{a}) = \mathbf{c} \cdot (\mathbf{a} \times \mathbf{b})$
- Parallelepiped volume: $V = |\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})|$

Right-Hand Rule and Signed Volume

- **Right-hand rule**: Curl fingers from $\mathbf{a}$ to $\mathbf{b}$, thumb points in direction of $\mathbf{a} \times \mathbf{b}$
- **Signed volume**: $V = \mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})$ is a signed scalar
  - $V > 0$: $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ form a right-handed system
  - $V < 0$: $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ form a left-handed system
  - $V = 0$: the three vectors are coplanar
- **Geometric interpretation**:
  - $\mathbf{b} \times \mathbf{c}$: normal vector of base, magnitude = base area
  - $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})$: projection of $\mathbf{a}$ onto normal $\times$ base area = volume
- **Actual volume**: $|V| = |\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})|$ (take absolute value)

Quaternion Basics

- A quaternion is written as $\mathbf{q} = [w,\, \mathbf{v}]^T = w + v_x\mathbf{i} + v_y\mathbf{j} + v_z\mathbf{k}$
- **Conjugate**: $\mathbf{q}^* = [w,\, -\mathbf{v}]^T$, **Inverse**: $\mathbf{q}^{-1} = \frac{\mathbf{q}^*}{\|\mathbf{q}\|^2}$
- **Hamilton product** (composition): if $\mathbf{q}_1 = [w_1,\, \mathbf{v}_1]^T$, $\mathbf{q}_2 = [w_2,\, \mathbf{v}_2]^T$,
  $\mathbf{q}_1 \otimes \mathbf{q}_2 = \left[w_1 w_2 - \mathbf{v}_1^T\mathbf{v}_2,\, w_1\mathbf{v}_2 + w_2\mathbf{v}_1 + \mathbf{v}_1 \times \mathbf{v}_2\right]^T$
- Multiplication is **non-commutative**: $\mathbf{q}_1 \otimes \mathbf{q}_2 \neq \mathbf{q}_2 \otimes \mathbf{q}_1$
- Axis-angle to quaternion: for axis $\mathbf{u}$ ($\|\mathbf{u}\| = 1$), angle θ,
  $\mathbf{q} = \left[\cos\frac{\theta}{2},\, \mathbf{u}\sin\frac{\theta}{2}\right]^T$
- Rotate vector $\mathbf{v}$ by unit quaternion: $\mathbf{v}' = \mathbf{q} \otimes [0, \mathbf{v}] \otimes \mathbf{q}^*$

Quaternion Basics

- Let $\mathbf{q} = [w,\ \mathbf{v}]^T$, with $[\mathbf{v}]_\times = \begin{bmatrix} 0 & -v_z & v_y \\ v_z & 0 & -v_x \\ -v_y & v_x & 0 \end{bmatrix}$

- **Left multiplication matrix**

$$L(\mathbf{q}) = \begin{bmatrix} w & -\mathbf{v}^T \\ \mathbf{v} & w\mathbf{I}_3 + [\mathbf{v}]_\times \end{bmatrix} = \begin{bmatrix} w & -v_x & -v_y & -v_z \\ v_x & w & -v_z & v_y \\ v_y & v_z & w & -v_x \\ v_z & -v_y & v_x & w \end{bmatrix}$$

- **Right multiplication matrix**

$$R(\mathbf{q}) = \begin{bmatrix} w & -\mathbf{v}^T \\ \mathbf{v} & w\mathbf{I}_3 - [\mathbf{v}]_\times \end{bmatrix} = \begin{bmatrix} w & -v_x & -v_y & -v_z \\ v_x & w & v_z & -v_y \\ v_y & -v_z & w & v_x \\ v_z & v_y & -v_x & w \end{bmatrix}$$

- For any quaternion $\mathbf{p}$: $\mathbf{q} \otimes \mathbf{p} = L(\mathbf{q})\mathbf{p}, \quad \mathbf{p} \otimes \mathbf{q} = R(\mathbf{q})\mathbf{p}$

Quaternion Basics

- Let angular velocity be $\omega \in \mathbb{R}^3$, and pure quaternion $\omega_q = [0,\, \omega]^T$
- **Quaternion kinematics** (body-frame angular velocity ω^B):

$$\dot{\mathbf{q}} = \frac{1}{2}\,\mathbf{q} \otimes [0, \omega^B] = \frac{1}{2} L(\mathbf{q}) \begin{bmatrix} 0 \\ \omega^B \end{bmatrix}$$

- Equivalent form (inertial-frame angular velocity ω^I):

$$\dot{\mathbf{q}} = \frac{1}{2}[0, \omega^I] \otimes \mathbf{q} = \frac{1}{2} R(\mathbf{q}) \begin{bmatrix} 0 \\ \omega^I \end{bmatrix}$$

- Recover angular velocity from $\mathbf{q}, \dot{\mathbf{q}}$:

$$[0, \omega^B] = 2\,\mathbf{q}^* \otimes \dot{\mathbf{q}}, \qquad [0, \omega^I] = 2\,\dot{\mathbf{q}} \otimes \mathbf{q}^*$$

- **Note**: choose one frame convention and keep it consistent in the whole derivation.

Quaternion Basics

- **Error quaternion** between desired attitude $\mathbf{q}_d$ and current attitude $\mathbf{q}$:

$$\mathbf{q}_e = \mathbf{q}_d \otimes \mathbf{q}^{-1} = \mathbf{q}_d \otimes \mathbf{q}^* \quad (\|\mathbf{q}\| = 1)$$

- Interpretation: if $\mathbf{q}_e = [w_e, \mathbf{v}_e]^T = \left[\cos\frac{\theta_e}{2}, \mathbf{u}_e \sin\frac{\theta_e}{2}\right]^T$, then $(\mathbf{u}_e, \theta_e)$ is the rotation from current to desired attitude

- **Shortest-arc convention**: because $\mathbf{q}$ and $-\mathbf{q}$ represent same rotation, enforce $w_e \geq 0$ (or flip sign when $w_e < 0$)

- Common control error vector:

$$\mathbf{e}_R = 2\,\mathrm{sgn}(w_e)\,\mathbf{v}_e \;\approx\; \theta_e \mathbf{u}_e \quad (\theta_e \text{ small})$$

- Left/right invariant forms (depending on convention):

$$\mathbf{q}_e^R = \mathbf{q}_d \otimes \mathbf{q}^{-1}, \qquad \mathbf{q}_e^L = \mathbf{q}^{-1} \otimes \mathbf{q}_d$$

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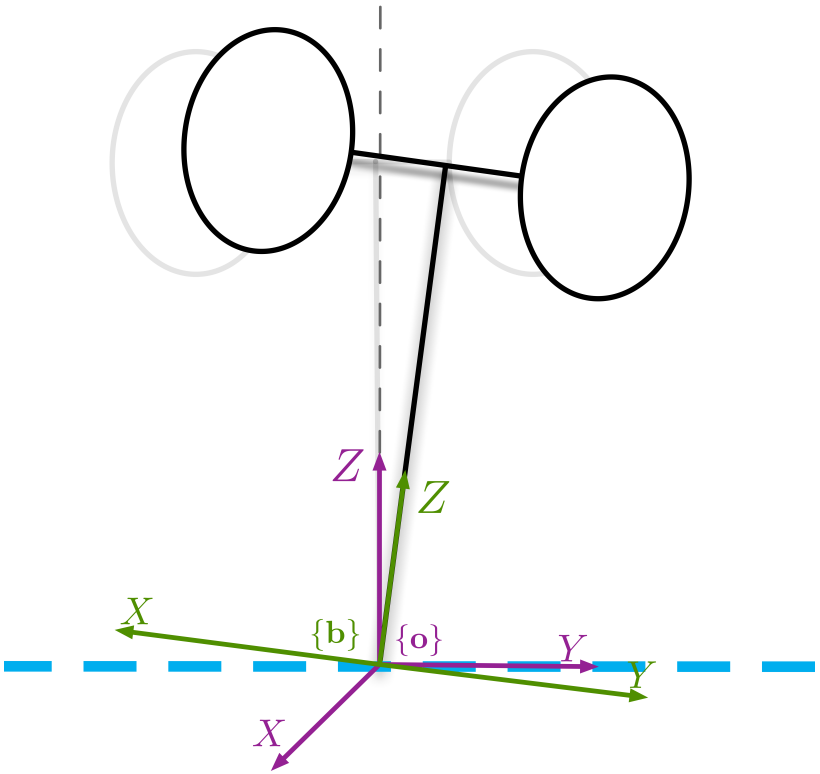
Model Setup and Assumptions



- Reaction wheel installation bias is ignored: $\alpha = 0$
- Yaw is ignored: $\psi = 0,\ \dot{\psi} = 0$
- Attitude states: roll ϕ, pitch θ
- Platform + wheel mass: m_p
- COM in body frame: ${}^{\mathbf{b}}M_p = [0, 0, l]^T$
- Gravity in world frame: ${}^{\mathbf{o}}g_p = [0, 0, -m_p g]^T$
- Wheel relative rates: ${}^{\mathbf{b}}\omega_w = [p_{w1}, p_{w2}, 0]^T$

Kinematics and Inertia

$$
{}^{\mathbf{o}}R_{\mathbf{b}} = R_y(\theta)R_x(\phi) = \begin{bmatrix} \cos\theta & \sin\theta\sin\phi & \sin\theta\cos\phi \\ 0 & \cos\phi & -\sin\phi \\ -\sin\theta & \cos\theta\sin\phi & \cos\theta\cos\phi \end{bmatrix}
$$

$$
{}^{\mathbf{b}}T_{\mathbf{o}}(\phi,\theta) = \begin{bmatrix} 1 & 0 & -\sin\theta \\ 0 & \cos\phi & \sin\phi\cos\theta \\ 0 & -\sin\phi & \cos\phi\cos\theta \end{bmatrix}, \quad {}^{\mathbf{o}}\dot{E}_p = \begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ 0 \end{bmatrix}
$$

$$
{}^{\mathbf{b}}\omega_p = {}^{\mathbf{b}}T_{\mathbf{o}}\,{}^{\mathbf{o}}\dot{E}_p = \begin{bmatrix} \dot{\phi} \\ \dot{\theta}\cos\phi \\ -\dot{\theta}\sin\phi \end{bmatrix}
$$

$$
{}^{\mathbf{b}}_{\mathbf{o}}I_p = \mathrm{diag}(J_{p1}, J_{p2}, J_{p3}), \quad {}^{\mathbf{b}}I_w = \mathrm{diag}(I_{w1}, I_{w1}, 0)
$$

Lagrangian

$$K_p = \frac{1}{2}\,{}^{\mathbf{b}}\omega_p^T\,{}^{\mathbf{b}}_{\mathbf{o}}I_p\,{}^{\mathbf{b}}\omega_p$$

$${}^{\mathbf{b}}\omega_{w,\text{total}} = {}^{\mathbf{b}}\omega_p + {}^{\mathbf{b}}\omega_w = \begin{bmatrix} \dot{\phi} + p_{w1} \\ \dot{\theta}\cos\phi + p_{w2} \\ -\dot{\theta}\sin\phi \end{bmatrix}$$

$$K_w = \frac{1}{2}\,{}^{\mathbf{b}}\omega_{w,\text{total}}^T\,{}^{\mathbf{b}}I_w\,{}^{\mathbf{b}}\omega_{w,\text{total}}$$

$$P = {}^{\mathbf{o}}g_p^T\,{}^{\mathbf{o}}R_{\mathbf{b}}\,{}^{\mathbf{b}}M_p = -m_p g l\cos\phi\cos\theta$$

$$L = K_p + K_w - P$$

Euler-Lagrange Equations

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) - \frac{\partial L}{\partial q_i} = Q_i$$

- Platform coordinates:

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\phi}}\right) - \frac{\partial L}{\partial \phi} = 0, \quad \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}}\right) - \frac{\partial L}{\partial \theta} = 0$$

- Wheel coordinates:

$$\frac{d}{dt}\left(\frac{\partial L}{\partial p_{w1}}\right) = \tau_1, \quad \frac{d}{dt}\left(\frac{\partial L}{\partial p_{w2}}\right) = \tau_2$$

- Gravity terms are already included in L via P.

Final Dynamics I (Platform, No Yaw)

$$
\begin{aligned}
&\frac{I_{w1}\dot{\theta}^2 \sin(2\phi)}{2} + I_{w1}\dot{\theta}p_{w2}\sin\phi + I_{w1}\dot{p}_{w1} \\
&+ \frac{J_{p2}\dot{\theta}^2 \sin(2\phi)}{2} - \frac{J_{p3}\dot{\theta}^2 \sin(2\phi)}{2} + (I_{w1} + J_{p1})\ddot{\phi} \\
&+ glm_p \sin\phi\cos\theta = 0
\end{aligned}
$$

$$
\begin{aligned}
&I_{w1}\dot{p}_{w2}\cos\phi + \ddot{\theta}\Big(I_{w1}\cos^2\phi + J_{p2}\cos^2\phi + J_{p3}\sin^2\phi\Big) \\
&- \dot{\phi}\Big(2I_{w1}\dot{\theta}\cos\phi + I_{w1}p_{w2}\Big)\sin\phi \\
&- \dot{\phi}\Big(2J_{p2}\dot{\theta}\cos\phi - 2J_{p3}\dot{\theta}\cos\phi\Big)\sin\phi \\
&+ glm_p \sin\theta\cos\phi = 0
\end{aligned}
$$

Final Dynamics II (Wheel, No Yaw)

$$I_{w1}\left(\ddot{\phi}+\dot{p}_{w1}\right)=\tau_1$$
$$I_{w1}(\ddot{\theta}\cos\phi-\dot{\phi}\dot{\theta}\sin\phi+\dot{p}_{w2})=\tau_2$$

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Adaptive Control Method

考虑二阶通道

$$\ddot{q} = f(q, \dot{q}) + bu$$

若 f 可线性参数化为 $f = Y(q, \dot{q}, \dot{q}_d, \ddot{q}_d)\Theta$，定义跟踪误差

$$e = q - q_d, \qquad s = \dot{e} + \lambda e$$

一种典型自适应控制律为

$$u = \frac{1}{b}\left(-Y\hat{\Theta} - k_s s + \ddot{q}_d - \lambda \dot{e}\right)$$

参数更新律取为

$$\dot{\hat{\Theta}} = \Gamma Y^T s$$

- Y: 回归矩阵，Θ: 未知参数，$\hat{\Theta}$: 参数估计
- $\Gamma \succ 0$: 自适应增益，$k_s > 0$: 误差收敛增益
- 设计步骤: 建模参数化 $\rightarrow$ 定义滤波误差 s $\rightarrow$ 构造控制律 $\rightarrow$ 用 Lyapunov 设计更新律
- 在 RWIP 中，Θ 可包含惯量、质心偏差、等效重力项等不确定参数

Sliding Mode Control Method

对于同样的跟踪任务，先定义滑模面

$$s = \dot{e} + \lambda e$$

将控制律写成“等效项 + 切换项”

$$u = u_{eq} + u_{sw}$$

其中常见设计为

$$u_{eq} = \frac{1}{b}\left(-\hat{f} + \ddot{q}_d - \lambda \dot{e}\right), \qquad u_{sw} = -\frac{k}{b}\,\mathrm{sat}\left(\frac{s}{\phi}\right)$$

其中

$$\mathrm{sign}(s) = \begin{cases} 1, & s > 0 \\ 0, & s = 0 \\ -1, & s < 0 \end{cases} \qquad \mathrm{sat}\left(\frac{s}{\phi}\right) = \begin{cases} 1, & s > \phi \\ \dfrac{s}{\phi}, & |s| \le \phi \\ -1, & s < -\phi \end{cases}$$

- u_{eq} 负责名义跟踪，u_{sw} 负责压制有界扰动和建模误差
- 若写成 $u_{sw} = -\frac{k}{b}\mathrm{sign}(s)$，鲁棒性更强，但抖振更明显
- $\phi > 0$ 为边界层厚度，$\mathrm{sat}(s/\phi)$ 在边界层内用线性段替代切换项
- 设计步骤: 定义滑模面 $\rightarrow$ 估计名义模型 $\rightarrow$ 选切换增益 k 覆盖扰动上界
- 在 RWIP 中，滑模项特别适合处理摩擦、外部冲击和未建模耦合项

Observer Method

若并非所有状态可直接测量，可先设计观测器，再将估计值送入控制器。

对于线性形式

$$\dot{x} = Ax + Bu, \qquad y = Cx$$

Luenberger 观测器写为

$$\dot{\hat{x}} = A\hat{x} + Bu + L(y - C\hat{x})$$

若还希望估计总扰动 d，可扩展为 ESO 结构

$$\dot{\hat{x}} = A\hat{x} + Bu + E\hat{d} + L_x(y - C\hat{x}), \qquad \dot{\hat{d}} = L_d(y - C\hat{x})$$

- L, L_x, L_d 通过极点配置或带宽法选择
- 设计步骤: 选观测状态 $\rightarrow$ 建立误差动态 $\rightarrow$ 配置观测器极点
- 在 RWIP 中可估计角速度、扰动力矩、未测状态，再接入自适应控制或滑模控制
- 工程上常见组合: Observer + Adaptive, Observer + SMC, ESO + feedback linearization

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论文概览

论文: Tracking Control for Inertia Wheel Inverted Pendulum Based on Fully Actuated System Theory

作者: Haowen Liu, Shenjia Chen, Ping Li, Guangren Duan

- 研究对象: 动量轮倒立摆（IWIP）的轨迹跟踪控制
- 核心思路: 将欠驱动系统通过微分同胚变换改写成高阶全驱系统（FAS）
- 主要目标: 在非线性模型下实现比局部线性控制更稳定的跟踪性能
- 主要增强: 引入可行参考轨迹约束，并设计扰动观测器进行在线补偿

系统模型与误差定义

论文从 Euler-Lagrange 模型出发：

$$D(q)\ddot{q} + g(q) = u + d, \quad q = \begin{bmatrix} q_1 \\ q_2 \end{bmatrix}, \quad u = \begin{bmatrix} 0 \\ \tau \end{bmatrix}$$

其中 q_1 为摆杆角，q_2 为飞轮角，$d = [d_1, d_2]^T$ 为外扰。

论文采用两条假设：

- $q_1 \in (-\frac{\pi}{2}, \frac{\pi}{2})$，即摆杆始终位于上半平面
- 扰动变化缓慢，满足 $\dot{d} \approx 0$

定义跟踪误差

$$e_1 = q_{d1} - q_1, \qquad e_2 = q_{d2} - q_2,$$

用于后续构造 FAS 输出变量。

FAS 变量构造

论文定义新的全驱输出变量

$$y_1 = \mu e_1 + e_2, \qquad \mu = \frac{D_{11}}{D_{12}}.$$

其中

$$y_{d1} = \mu q_{d1} + q_{d2}, \qquad c_1 = (1-\mu)M, \qquad c_2 = b_1 - b_2.$$

在 $\dot{d}_1 = 0$ 的假设下，原论文中的导数链可写为

$$\begin{aligned}
y_1 &= y_{d1} - (\mu q_1 + q_2), \\
\dot{y}_1 &= \dot{y}_{d1} - (\mu \dot{q}_1 + \dot{q}_2), \\
\ddot{y}_1 &= \ddot{y}_{d1} + c_1 \sin(q_1) + c_2 d_1, \\
y_1^{(3)} &= y_{d1}^{(3)} + c_1 \cos(q_1)\dot{q}_1, \\
y_1^{(4)} &= y_{d1}^{(4)} + c_1 \left[-\sin(q_1)\dot{q}_1^2 + \cos(q_1)\ddot{q}_1\right].
\end{aligned}$$

通过消去 $q_1, \dot{q}_1$，原欠驱动系统可改写为四阶 SISO FAS：

$$y_1^{(4)} = f(\ddot{y}_1, \dot{y}_1) + b(\ddot{y}_1)u_1.$$

带偏移的模型构造

为了和原论文结果做对比，额外考虑摆角存在常值偏移 Δ：

$$\ddot{q}_1 = M\sin(q_1+\Delta) - b_1\tau + b_1 d_1 - b_1 d_2, \quad \ddot{q}_2 = -M\sin(q_1+\Delta) + b_2\tau - b_1 d_1 + b_2 d_2.$$

此时误差与 FAS 输出改写为

$$e_1 = q_{d1} - (q_1 + \Delta), \qquad e_2 = q_{d2} - q_2, \qquad y_1 = \mu e_1 + e_2.$$

若 Δ 为常值，则

$$\sin(q_1 + \Delta) = \sin q_1 \cos\Delta + \cos q_1 \sin\Delta,$$

因而原论文中所有由 $\sin q_1, \cos q_1$ 组成的重力项，都需要整体替换为带偏移角的形式。

带偏移的 y_1 导数链

利用 $\mu = b_2/b_1$ 消去输入项，并采用 $\dot{d}_1 = 0$、$\dot{\Delta} = 0$，有

$$
\begin{aligned}
y_1 &= y_{d1} - (\mu(q_1 + \Delta) + q_2), \\
\dot{y}_1 &= \dot{y}_{d1} - (\mu\dot{q}_1 + \dot{q}_2), \\
\ddot{y}_1 &= \ddot{y}_{d1} + c_1 \sin(q_1 + \Delta) + c_2 d_1, \\
y_1^{(3)} &= y_{d1}^{(3)} + c_1 \cos(q_1 + \Delta)\dot{q}_1, \\
y_1^{(4)} &= y_{d1}^{(4)} + c_1 \left[-\sin(q_1 + \Delta)\dot{q}_1^2 + \cos(q_1 + \Delta)\ddot{q}_1\right].
\end{aligned}
$$

通过消去 $q_1, \dot{q}_1$，原欠驱动系统可改写为四阶 SISO FAS：

$$
y_1^{(4)} = f(\ddot{y}_1, \dot{y}_1) + b(\ddot{y}_1)u_1.
$$

无偏移与有偏移的对比

原论文

$$e_1 = q_{d1} - q_1$$

$$\ddot{y}_1 = \ddot{y}_{d1} + c_1 \sin(q_1) + c_2 d_1$$

$$y_1^{(3)} = y_{d1}^{(3)} + c_1 \cos(q_1)\dot{q}_1$$

$$y_1^{(4)} = y_{d1}^{(4)} + c_1\left[-\sin(q_1)\dot{q}_1^2 + \cos(q_1)\ddot{q}_1\right]$$

考虑偏移 Δ

$$e_1 = q_{d1} - (q_1 + \Delta)$$

$$\ddot{y}_1 = \ddot{y}_{d1} + c_1 \sin(q_1 + \Delta) + c_2 d_1$$

$$y_1^{(3)} = y_{d1}^{(3)} + c_1 \cos(q_1 + \Delta)\dot{q}_1$$

$$y_1^{(4)} = y_{d1}^{(4)} + c_1\left[-\sin(q_1 + \Delta)\dot{q}_1^2 + \cos(q_1 + \Delta)\ddot{q}_1\right]$$

偏移不会改变 FAS 的设计框架，但会改变误差定义和所有重力非线性项。

名义 FAS 控制律设计

原文先将 IWIP 写成四阶 FAS

$$y_1^{(4)} = f(\ddot{y}_1, \dot{y}_1) + b(\ddot{y}_1)u_1.$$

为了指定期望误差动力学，选取名义控制律

$$u_1 = -\frac{1}{b(\ddot{y}_1)}\left[f(\ddot{y}_1, \dot{y}_1) + \sum_{i=0}^{3} K_i y_1^{(i)}\right].$$

代回 FAS 可得闭环

$$y_1^{(4)} + K_3 y_1^{(3)} + K_2 \ddot{y}_1 + K_1 \dot{y}_1 + K_0 y_1 = 0.$$

因此特征多项式取为

$$\lambda_c(s) = s^4 + K_3 s^3 + K_2 s^2 + K_1 s + K_0.$$

按原文采用极点配置

$$\lambda_c(s) = (s + \omega_c)^4 = s^4 + 4\omega_c s^3 + 6\omega_c^2 s^2 + 4\omega_c^3 s + \omega_c^4,$$

从而得到

$$K_0 = \omega_c^4, \qquad K_1 = 4\omega_c^3, \qquad K_2 = 6\omega_c^2, \qquad K_3 = 4\omega_c.$$

可行参考轨迹约束

原文指出该 FAS 实际上是 SISO 结构，因此 $q_{d1}(t), q_{d2}(t)$ 不能完全独立指定。当 $y_1 \to 0$ 且 $\ddot{y}_1 \to 0$ 时，有

$$y_1 = \mu e_1 + e_2 = 0, \qquad \ddot{y}_1 = \ddot{y}_{d1} + c_1 \sin(q_1) + c_2 d_1 = 0.$$

为保证零误差跟踪，期望轨迹必须满足

$$\ddot{y}_{d1} = -c_1 \sin(q_{d1}) - c_2 d_1 .$$

代回 $y_{d1} = \mu q_{d1} + q_{d2}$ 并利用参数定义，可写成论文中的约束式

$$D_{11}\ddot{q}_{d1} + D_{12}\ddot{q}_{d2} - \bar{m} g \sin(q_{d1}) = d_1 .$$

这个条件的物理含义是：

- 期望摆角和飞轮轨迹必须与系统动力学一致
- 参考输入的自由度被约束到一维可实现流形上
- 若不满足该约束，控制器只能做到有界误差而非严格零误差

带扰动补偿的实际控制律

将扰动估计并入控制后，原文给出的实际输入力矩为

$$\tau = \frac{1}{b_1 b(\ddot{y}_1)} \left[f(\ddot{y}_1, \dot{y}_1) + \sum_{i=0}^{3} K_i \hat{y}_1^{(i)} \right] + \frac{M}{b_1} \sin(q_1) + \hat{d}_1 - \hat{d}_2.$$

代回子 FAS 后得到受扰闭环

$$y_1^{(4)} + \sum_{i=0}^{3} K_i y_1^{(i)} + \Delta(t, y_1, \dot{y}_1, \ddot{y}_1) = 0,$$

其中扰动项为

$$\Delta = (b_1 \tilde{d}_1 - b_1 \tilde{d}_2) b(\ddot{y}_1) - K_2 \tilde{d}_1.$$

实际实现时，$\hat{d}$ 由扰动观测器在线给出，用于补偿约束失配和外部扰动。

考虑偏移 Δ 的名义控制律

若前面建立的 FAS 变量改为

$$e_1 = q_{d1} - (q_1 + \Delta), \qquad y_1 = \mu e_1 + e_2,$$

且 Δ 为常值，则导数链中的重力项全部替换为

$$\sin(q_1 + \Delta), \qquad \cos(q_1 + \Delta).$$

此时四阶 FAS 仍写成同一结构

$$y_1^{(4)} = f_\Delta(\ddot{y}_1, \dot{y}_1) + b_\Delta(\ddot{y}_1) u_{1,\Delta}.$$

因而名义控制律可以平行写为

$$u_{1,\Delta} = -\frac{1}{b_\Delta(\ddot{y}_1)} \left[f_\Delta(\ddot{y}_1, \dot{y}_1) + \sum_{i=0}^{3} K_i y_1^{(i)} \right].$$

对应闭环仍然是

$$y_1^{(4)} + K_3 y_1^{(3)} + K_2 \ddot{y}_1 + K_1 \dot{y}_1 + K_0 y_1 = 0,$$

所以极点配置不需要改变，仍取

$$\lambda_c(s) = (s + \omega_c)^4, \quad K_0 = \omega_c^4,\ K_1 = 4\omega_c^3,\ K_2 = 6\omega_c^2,\ K_3 = 4\omega_c.$$

考虑偏移 Δ 的实际力矩控制律

由

$$\ddot{y}_1 = \ddot{y}_{d1} + c_1 \sin(q_1 + \Delta) + c_2 d_1$$

可见在执行层，原文中的补偿项

$$\frac{M}{b_1} \sin(q_1)$$

应同步改为

$$\frac{M}{b_1} \sin(q_1 + \Delta).$$

因此带扰动观测器的实际输入力矩可写成

$$\tau_\Delta = \frac{1}{b_1 b_\Delta(\ddot{y}_1)} \left[f_\Delta(\ddot{y}_1, \dot{y}_1) + \sum_{i=0}^{3} K_i \hat{y}_1^{(i)} \right] + \frac{M}{b_1} \sin(q_1 + \Delta) + \hat{d}_1 - \hat{d}_2.$$

若进一步展开偏移角，则

$$\tau_\Delta = \frac{1}{b_1 b_\Delta(\ddot{y}_1)} \left[f_\Delta + \sum_{i=0}^{3} K_i \hat{y}_1^{(i)} \right] + \frac{M}{b_1} (\sin q_1 \cos \Delta + \cos q_1 \sin \Delta) + \hat{d}_1 - \hat{d}_2.$$

这说明偏移不会改变控制器设计框架，只会改变：

- FAS 映射中的非线性函数 f, b
- 力矩控制中的重力补偿项
- 可行参考轨迹约束中的正弦项

论文结论与可借鉴点

- 论文的核心贡献不是重新推导 IWIP 模型，而是把 IWIP 提升为可直接设计的四阶 FAS
- 参考轨迹约束揭示了欠驱动系统只能跟踪**可实现**的目标，而非两个完全独立的输出
- 扰动观测器补偿后，仿真与实验都表明跟踪性能和鲁棒性得到改善
- 对你当前这份 beamer 而言，这篇论文最值得保留的主线是：“原系统建模 $\rightarrow$ FAS 变换 $\rightarrow$ 四阶控制器 $\rightarrow$ 轨迹约束 $\rightarrow$ 扰动观测器”

1 Prerequisites

2 RWIP Lagrangian Modeling

3 Control and Observer Methods

4 Paper Summary: FAS-Based IWIP Tracking

5 References

Thanks!